Database and Networking

10. Develop an Android application implementing SQLite database CRUD operations.

ProjectName: SQLLiteCRUDDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <EditText

       android:id="@+id/etName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter Name"/>

   <EditText

       android:id="@+id/etNewName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter New Name"/>

   <Button

       android:id="@+id/btnInsert"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Insert"/>

    <Button

        android:id="@+id/btnView"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="View Data"/>

    <Button

        android:id="@+id/btnUpdate"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Update"/>

    <Button

        android:id="@+id/btnDelete"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Delete"/>

    <TextView

        android:id="@+id/txtResult" android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:textSize="18sp"

         android:paddingTop="20dp"/>

</LinearLayout>
```

Create a Database Helper Class.

Right Click:

java package → New → Java Class

Class Name: DBHelper

```
package com.example.sqllitecruddemo;


import android.content.ContentValues;

import android.content.Context;

import android.database.Cursor;

import android.database.sqlite.SQLiteDatabase;

import android.database.sqlite.SQLiteOpenHelper;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {

        super(context, "StudentDB", null, 1);

    }


    @Override

    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE student(name TEXT)");

    }


    @Override

    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        db.execSQL("DROP TABLE IF EXISTS student");
```

```
        onCreate(db);
    }


    // Insert
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();


        ContentValues values = new ContentValues();
        values.put("name", name);


        db.insert("student", null, values);
    }


    // View
    public String viewData() {
        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT * FROM student", null);


        StringBuilder data = new StringBuilder();


        while (cursor.moveToNext()) {
            data.append(cursor.getString(0)).append("\n");
        }


        cursor.close();
```

```
    return data.toString();
}

// Update
public void updateData(String oldName, String newName) {
    SQLiteDatabase db = this.getWritableDatabase();

    ContentValues values = new ContentValues();
    values.put("name", newName);

    db.update(
            "student",
            values,
            "name=?",
            new String[]{oldName}
    );
}

// Delete
public void deleteData(String name) {
    SQLiteDatabase db = this.getWritableDatabase();

    db.delete(
            "student",
            "name=?",
```

```
                new String[]{name}

        );

    }

}
```

MainActivity.java

```
package com.example.sqllitecruddemo;



import android.os.Bundle;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;



import androidx.appcompat.app.AppCompatActivity;



public class MainActivity extends AppCompatActivity {



    EditText etName, etNewName;

    TextView txtResult;



    Button btnInsert, btnView, btnUpdate, btnDelete;



    DBHelper dbHelper;
```

```java
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etName = findViewById(R.id.etName);
    etNewName = findViewById(R.id.etNewName);
    txtResult = findViewById(R.id.txtResult);

    btnInsert = findViewById(R.id.btnInsert);
    btnView = findViewById(R.id.btnView);
    btnUpdate = findViewById(R.id.btnUpdate);
    btnDelete = findViewById(R.id.btnDelete);

    dbHelper = new DBHelper(this);

    // Insert
    btnInsert.setOnClickListener(v -> {

        String name = etName.getText().toString();

        dbHelper.insertData(name);

        Toast.makeText(
                this,
                "Data Inserted",
```

```
                Toast.LENGTH_SHORT
        ).show();
    });

    // View
    btnView.setOnClickListener(v -> {

        String data = dbHelper.viewData();

        txtResult.setText(data);
    });

    // Update
    btnUpdate.setOnClickListener(v -> {

        String oldName = etName.getText().toString();
        String newName = etNewName.getText().toString();

        dbHelper.updateData(oldName, newName);

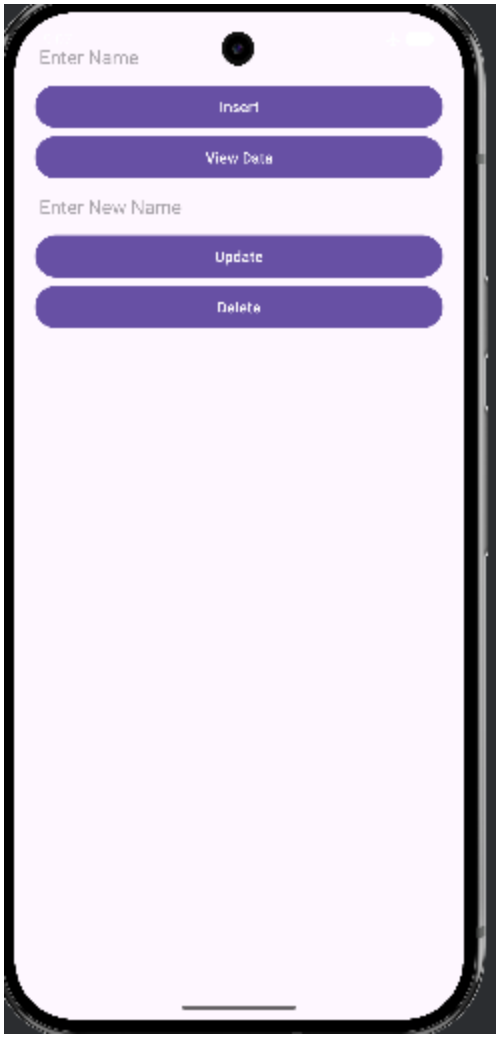
        Toast.makeText(
                this,
                "Data Updated",
                Toast.LENGTH_SHORT
        ).show();
    });
```

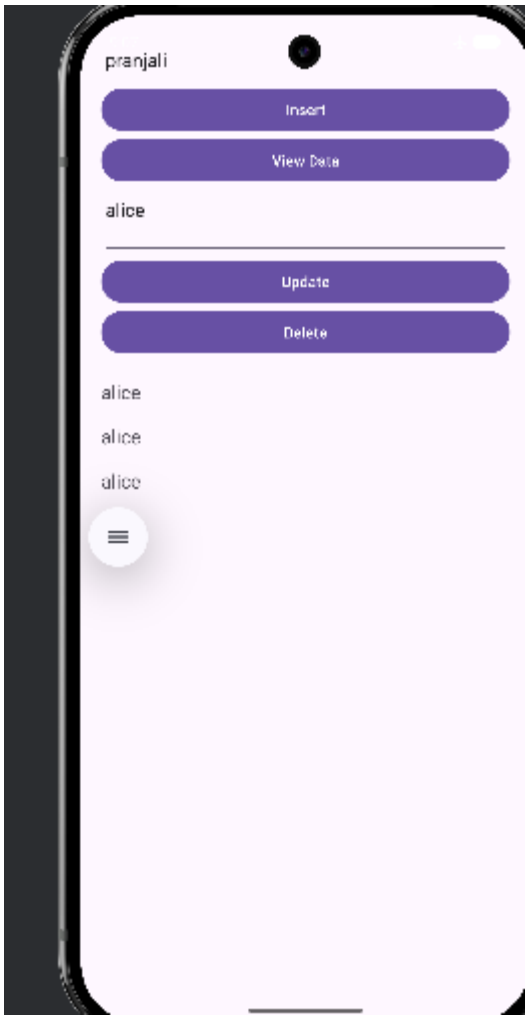
```
        // Delete
        btnDelete.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.deleteData(name);

            Toast.makeText(
                    this,
                    "Data Deleted",
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


Enter Name
Insert
View Data
Enter New Name
Update
Delete


pranjali
Insert
View Data
alice
Update
Delete
alice
alice
alice